



| Course Syllabus | AA-5300 - 12 | Advanced Analytics |
| Spring 2025-1 |

Instructor: Juan Ignacio Gonzalez, Ph.D.	Course Meeting Time: Tuesdays & Thursdays, 9:30 to 11:30 AM.
Email: juanignacio.gonzalez@slu.edu	Meeting Location: Litteken Hall, Room 101
Office Location: Brouster Hall, office 254	Pre-requisites / Co-requisites: AA 5000; (AA 5222 or AA 5223)
Office Hours: Wednesdays, 10:00 to 11:00 AM.	Course Assistant: N/A

COURSE CATALOG DESCRIPTION

This course covers several commonly used advanced analytical methods involving statistical learning. Applications of these methods on datasets drawn from several fields will be emphasized, alongside a coverage of visualizations of data and results. Students will also learn how to automate tasks in various phases of statistical analyses and in creating useful visualizations of data and results.

COURSE GOALS

After successfully completing this course, the student will be able to:

1. Understand the key concepts underlying commonly used statistical learning techniques.
2. Apply the concepts associated with automating various tasks in distinct phases of a data analysis project.
3. Evaluate the effectiveness of various statistical learning techniques within the context of a given dataset.
4. Present the results of applying statistical learning in formats that aid decision-making.

REQUIRED MATERIALS

There are two versions of the textbook for this course: a Python one and an R one. The concepts covered are the same; they differ in the code examples, using Python and R, respectively.

- **R-based:** James, G., Witten, D., Hastie, T., and Tibshirani, R. (2023). *An Introduction to Statistical Learning with Applications in R*. Second Edition. Springer.
- **Python-based:** James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J. (2023). *An Introduction to Statistical Learning with Applications in Python*. First Edition. Springer.



You can get both editions here: <https://www.statlearning.com/>.

- **Software:** R with R Studio and Python (either via Anaconda, locally installed, - Jupyter-, or Google Colab).

TIME EXPECTATIONS AND GRADING

Students should be prepared to spend between 10-15 hours per week on work for this course. The course's fast-paced nature demands that students stay on top of work and communicate with the instructor as soon as any problems or concerns arise.

COURSE DELIVERY/METHODOLOGY

This course is delivered on-ground. Students registered in this course will be responsible for participating in the course in the following ways:

- **Class attendance.** Students are required to attend scheduled classes to enhance knowledge and skill acquisition throughout the term.
- **Student Introduction.** The "Introduce Yourself" discussion post will be used to give your professor and your classmates a little background information about yourself. This discussion post will also be used to verify your attendance in the course. *For this reason, it is mandatory for you to complete them by the dates and hours shown on the Course Calendar.*
- **Readings.** Assigned readings should be read at least once, with multiple readings recommended for difficult material. Optional reference documents are available for this course, your job, your capstone project, or work in the Analytics profession.
- **In-class presentations.** Teams will present solutions to mini-cases weekly, reinforcing models, concepts, and practices. Presentations will be peer-reviewed by classmates and graded in class, on a form provided by the instructor. Depending on the number of teams formed (3-4 members are recommended), the instructor will ensure that each team presents at least once during the term. *This is a graded activity, and points will be awarded only to students who attend the session and actively cooperate in solving this class challenge.*
- **Self-assessments.** Self-assessments are quick tests with up to six multiple-choice questions in Canvas, based on in-class exercises and prior activities (readings, videos). *These assessments, conducted at the end of the first session of each week, count toward the final grade and are credited only to students who attend the classes in which they are administered. **Remote completion of self-assessments is prohibited.***
- **Mid-term and final evaluations.** These summative evaluations assess the student's comprehension, application, and synthesis of knowledge acquired across the term. This evaluation consists of an oral presentation.



- **Final Project:** A final workbook project consisting of a case study based on a dataset, selected by the student. This evaluation consists of a video oral presentation.

All activities are to be submitted individually, except for the in-class presentations.

COURSE GRADING SCALE

Component	Points	Total	%	Letter Grade Assignment
Self-assessments (7)	7x4	28	13%	94 to 100 = A
Mid-term evaluation (1)	1x40	40	18%	90 to 93 = A-
Final exam (1)	1x40	40	18%	87 to 89 = B+
In-class presentations (6)	6x5	30	13%	84 to 86 = B
(80% corresponds to the grade assigned to submission and 20% comes from the individual evaluation of team members)				80 to 83 = B-
Reflection essay	1x5	5	2%	77 to 79 = C+
Final Project Data Analysis		80	36%	70 to 76 = C
<i>Intermediate deliverables (3)</i>	3x10	30	14%	60 to 69 = D
<i>Final deliverable (1) (report and video)</i>	1x50	50	22%	0 to 59 = F

TOTAL	223	100%
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FQ - The grade of FQ should be given to those students who had ceased attending/participating in a course and as a result earned a failing grade.

LATE OR MISSING ASSIGNMENTS

Please refer to the course calendar for the due dates of assignments. Assignments are due by 11:59 PM on the week and day listed in the calendar and instructions (except for the mid-term and final evaluations, as well as the in-class presentations and self-assessments, which will be submitted at the end of the corresponding class). Any submission made after this time will be considered late. Late assignments will incur a



penalty of a 4.17% deduction for each hour past the due date. Students are not required to obtain special permission or authorization to submit deliverables past the due date, but such submissions will be subject to this Late Submission Policy.

ATTENDANCE POLICY

Success in this course depends on your attendance, and missing a full week of class, whether one or multiple sessions, during an eight-week term can seriously impact your learning. Failure to attend two weeks or more puts you at serious risk of failing the course. Medical documentation may be necessary in cases of serious illness or medical conditions which impact your ability to participate in the course for an extended period. The instructor reserves the right to excuse absences, and/or accept late work. In the event of a chronic medical condition or disability, students should contact the Dean of Students and/or the Office of Disability Services to explore potential accommodation.

EARLY LEAVING POLICY

As per the nature of our on-ground course, no student is permitted to complete any content online. Attendance in class is mandatory, not only due to syllabus rules but also to comply with immigration policies (requirements for F-1 visa holders) in the case of international students.

The course instructor may consider a force majeure cause as a valid reason for a student's absence from class or early departure from the term. Such causes must relate to unplanned, unforeseen personal or family circumstances that could significantly affect the student's physical and mental integrity. These circumstances will be evaluated by the instructor, who will review all relevant and official documentation provided. Appropriate accommodation and permissions will be granted, if applicable, once the case is assessed and a response is issued. Seeking cheaper airfares, attending family or friends' reunions, engaging in social commitments, or scheduling planned medical treatments or appointments are examples of invalid reasons for class absences or requests for early departures.

RE-EVALUATION OF ACTIVITIES

Students may require re-evaluation of graded course activities. These requests must be made by posting in the discussion group named "**Re-evaluation of Grades Request**," located at the bottom of the **Modules section** in Canvas. *Please follow the specific instructions provided there for posting your request. Any other requests on this matter submitted to the instructor via another method (email, Canvas message) will not be addressed.*

ACADEMIC INTEGRITY

Academic integrity is the commitment to and demonstration of honest and moral behavior in an academic setting. Since the mission of the University is "the pursuit of truth for the greater glory of God and for the service of humanity," acts of integrity are essential to its very reason for existence. Thus, the University regards academic integrity as a matter of



serious importance. Academic integrity is the foundation of the academic assessment process, which in turn sustains the ability of the University to certify to the outside world the skills and attainments of its graduates. Adhering to the standards of academic integrity allows all members of the University to contribute to a just and equitable learning environment that cultivates moral character and self-respect. The full University-level Academic Integrity Policy can be found here:

<https://www.slu.edu/provost/policies/academic-and-course/academic-integrity-policy.pdf>

ARTIFICIAL INTELLIGENCE TOOLS

I encourage the use of generative AI tools to assist in starting research activities, guiding knowledge search, and providing support. While AI is a powerful tool to enhance research efficiency and broaden perspectives, it should complement—not replace—your personal work, creativity, and intellectual development. Any evidence of copy-paste practices will be addressed accordingly.

STUDENT SUCCESS CENTER

The Student Success Center (SSC) supports students in reaching their goals in and out of the classroom. Providing a variety of resources, the Student Success Center houses both the Center for Accessibility and Disability Resources (CADR) and Academic Support, which includes Tutoring, Supplemental Instruction, University Writing Services, and Student Success Coaching. The Student Success Center can be accessed by the Busch Student Center, Suite 331, and students can make an appointment with any SSC resource via EAB Navigate. To learn more about the Student Success Center and its resources, please visit our website: <https://www.slu.edu/life-at-slu/student-success-center/index.php>

DISABILITY ACCOMMODATIONS

Students with a documented disability who wish to request academic accommodation must formally register their disability with the University. Once successfully registered, students also must notify their course instructor that they wish to use their approved accommodations in the course.

Please contact the Center for Accessibility and Disability Resources (CADR) to schedule an appointment to discuss accommodation requests and eligibility requirements. Most students on the St. Louis campus will contact CADR, located in the Student Success Center and available by email at accessibility_disability@slu.edu or by phone at 314-977-3484. Once approved, information about a student's eligibility for academic accommodation will be shared with course instructors by email from CADR and within the instructor's official course roster. Students who do not have a documented disability, but who think they may have one, also are encouraged to contact CADR. Confidentiality will be observed in all inquiries.

TITLE IX



Saint Louis University and its faculty are committed to supporting our students and seeking an environment that is free of bias, discrimination, and harassment. If you have encountered any form of discrimination based on sex, including sexual harassment, sexual assault, stalking, domestic or dating violence, we encourage you to report this to the University. Discrimination based on sex includes discrimination based on assigned sex at birth, sex characteristics, pregnancy and pregnancy related conditions, sexual orientation, and gender identity. If you speak with a faculty member about an incident that involves a Title IX matter, that faculty member must notify SLU's Title IX Coordinator that you shared an experience relating to Title IX. This is true even if you ask the faculty member not to disclose the incident. The Title IX Coordinator will then be available to assist you in understanding all your options and in connecting you with all resources on and off campus.

If you are pregnant or experiencing a pregnancy related condition, the Title IX Coordinator can assist you in understanding your rights and options as well as provide supportive measures.

Anna Kratky is the Title IX Coordinator at Saint Louis University (DuBourg Hall, room 36; anna.kratky@slu.edu; 314-977-3886). If you wish to speak with a confidential source, you may contact the counselors at the University Counseling Center at 314-977-TALK or make an anonymous report through SLU's Integrity Hotline by calling 1-877-525-5669 or online at <http://www.lighthouse-services.com/slu>. To view SLU's policies, and for resources, visit this [site](#).

WELLNESS

With our Jesuit commitment to *cura personalis*, “caring for the whole person,” the University sees your academic success as connected to your health and well-being and provides resources to support your holistic wellness. All students experience stressors and challenges at some point, and seeking support is both normal and beneficial. Such challenges may be the result of academic concerns (such as those related to particular assignments or content in a course), or they may be more personal in nature (such as concerns related to relationships, mental health, medical issues, loss, identities, alcohol or drugs, housing or food security, finances, or local/world events, among other things). If you experience these or other difficulties impacting your well-being and/or academic work, please consider seeking support from the resources available to you.

UNIVERSITY COUNSELING CENTER

The University Counseling Center (UCC) offers free, short-term, purposeful counseling to Saint Louis University undergraduate and graduate students. UCC counselors are highly trained clinicians who can assist with a variety of issues, such as adjustment to college life, troubling changes in mood, and chronic psychological conditions. To make an appointment for a wellness consultation, call 314-977-8255 (TALK), or visit the clinic on the



second floor of Wuller Hall. For after-hours needs, please press #9 after dialing the clinic number.

BASIC NEEDS SECURITY

Students experiencing food insecurity, housing insecurity, and any other challenges that are impacting their personal and/or academic wellbeing are encouraged to contact the Dean of Student's Office for support. Students can submit an intake form, email deanofstudents@slu.edu, or call 314-977-9378 to connect with their office. Students may also communicate directly with their instructors about any challenges they are experiencing to receive support and resource referrals.

STUDENT COURSE RECORDING POLICY

Saint Louis University prohibits recording and transmission of classroom lectures and discussions by students unless written permission from the class instructor has been obtained and all students in the class as well as guest speakers have been informed that audio/video recording may occur. Recordings, course materials, and lecture notes may not be exchanged or distributed for commercial purposes, for compensation, or for any other purpose other than study by students enrolled in the class. Public distribution of such materials may constitute copyright infringement, violating federal or state law or University policy. Violation of this policy may subject a student to disciplinary action under the University's Student Handbook.



AA5300 Spring 2025-1. Calendar and Roadmap.

WK	SESSION	TOPICS	READINGS		EVALUATIONS	DISCUSSIONS	IN-CLASS PRESENTATIONS	FINAL PROJECT	OTHER ACTIVITIES
			Textbook and/or readings	DUE DATE (Readings)					
0	Pre-work	None	Chapters 1 & 2	13-Jan	-	-			AW0: Setting up R/Python and obtain the textbook(s). Due: 1/13
1	Regression-based models.	<i>Predictive modeling fundamentals:</i> 1.1 Regression via multiple linear regression 1.2 Classification via logistic regression	Chapters 3, 4 & 5	13-Jan	-	DW1: Introduce yourself in the discussion forum. Due: 1/20	INC1: In-class presentation 1. Due: 1/16	-	SA1: Respond to the self-assessment on first class of the week, at the end of the session.
2	Regularization and non-linear models	2.1 Regularization methods 2.2 Non-linear parametric models	Chapters 6 & 7	20-Jan	-	-	INC2: In-class presentation 2. Due: 1/23	FP1: Final Project Deliverable 1. Due: 1/27	SA2: Respond to the self-assessment on first class of the week, at the end of the session.
3	Three-based methods	3.1 Decision trees 3.2 Random forests 3.3 Boosted trees 3.4 Bayesian additive regression trees [optional]	Chapter 8	27-Jan	-	-	INC3: In-class presentation 3. Due: 1/28	-	SA3: Respond to the self-assessment on first class of the week, at the end of the session.
4	SVM and explainability	4.1 Support Vector Machines 4.2 Explainability of models	Chapter 9 & additional reading the instructor	3-Feb	MTE: Mid-Term Evaluation. Until: 2/19 for personal presentations	-	-	-	SA4: Respond to the self-assessment on first class of the week, at the end of the session.
5	Unsupervised learning	5.1 k-Means 5.2 Hierarchical Clustering 5.3 Density-Based Approaches	Chapter 12	10-Feb	-	-	INC5: In-class presentation 5. Due: 2/13	FP2: Final Project Deliverable 2. Due: 2/17	SA5: Respond to the self-assessment on first class of the week, at the end of the session.
6	Deep Learning	6.1 Artificial Neural networks	Chapter 10 & additional readings.	17-Feb	-	-	INC6: In-class presentation 6. Due: 2/20	-	SA6: Respond to the self-assessment on first class of the week, at the end of the session.
7	Assessing models	7.1 Model comparison/evaluation 7.2 Considerations for decision-making	Materials provided by the instructor.	24-Feb	-	-	INC7: In-class presentation 7. Due: 2/27	FP3: Final Project Deliverable 3. Due: 3/1	SA7: Respond to the self-assessment on first class of the week, at the end of the session.
8	Orientation and work on Final Project	No new topic will be offered	None	3-Mar	FE: Final Evaluation. Due: 3/6	-	-	FPR: Final Project Data Analysis. Due: 3/8 RE: Reflection Essay. Both: Due: 3/8	-
Subject to change, points on this Roadmap/Calendar supersede the syllabus. Your homework and activities are always due in Canvas by 11:59 PM on the due date marked in this calendar, or any update published or shared by the instructor, effectively communicated by a class announcement.									